Linear Programming

CS5491: Artificial Intelligence
ZHICHAO LU

Content Credits: Prof. Wei's CS4486 Course and Prof. Boddeti's AI Course

ANNOUNCEMENTS

Project proposal: 15 October 2026, 23:59 (BJT)
Submit through Canvas. The group project is worth 30% of the course grade.

Midterm: 22 October 2026
Worth 20% of the course grade. Check Canvas for preparation materials.

WHAT TO EAT?

We want to meet the given nutrition targets at minimum cost.

We can choose the amount of stir-fry (ounces) and boba (fluid ounces).

Nutrition targets

$2000 \leq \text{Calories} \leq 2500$

$\text{Sugar} \leq 100\,\text{g}$

$\text{Calcium} \geq 700\,\text{mg}$

Food	Cost (units)	Calories	Sugar (g)	Calcium (mg)
stir-fry (per oz)	**1**	**100**	**3**	**20**
Boba (per fl oz)	**0.4**	**50**	**4**	**70**

What is the cheapest way to meet these targets with this menu?

How much stir-fry (ounces) and boba (fluid ounces) should we buy?

CAN SEARCH ALGORITHMS HELP?

Essence of search algorithms:

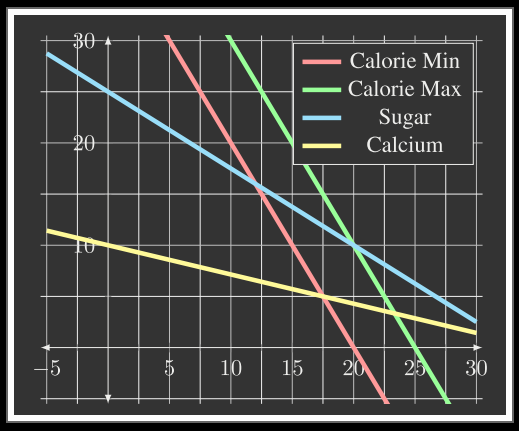
- explore candidate states until a goal is reached
- use additional information to avoid unnecessary search

Can we adopt a similar strategy here?

OPTIMIZATION

Problem Description

Graphical Representation



Optimization Representation

$$\begin{aligned} \min_{\mathbf{x}} && \mathbf{c}^T\mathbf{x} \\ s.t. && \mathbf{A}\mathbf{x} \leq \mathbf{b} \end{aligned}$$

WHAT TO EAT?: CONSTRAINTS

We want to meet the given nutrition targets at minimum cost.

We can choose the amount of stir-fry (ounces) and boba (fluid ounces).

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CONSTRAINT SATISFACTION PROBLEMS

Map Coloring $$ \begin{aligned} Any &~\mathbf{x} \\ s.t. &~\mathbf{x} {\rm{~satisfies~constraints}} \end{aligned} $$

WHAT TO EAT?: COSTS

We want to meet the given nutrition targets at minimum cost.

We can choose the amount of stir-fry (ounces) and boba (fluid ounces).

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OPTIMIZATION FORMULATION

x_1: stir-fry (oz); x_2: boba (fl oz). $\mathbf{x}\in\mathbb{R}_{\geq0}^2$.

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> Diet Problem $$ \begin{aligned} \min_{\mathbf{x}} && 1x_1 + 0.4x_2 \\ \text{s.t.} && 100x_1 + 50x_2 \geq 2000 \\ && 100x_1 + 50x_2 \leq 2500 \\ && 3x_1 + 4x_2 \leq 100 \\ && 20x_1 + 70x_2 \geq 700 \end{aligned} $$

OPTIMIZATION FORMULATION

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> Diet Problem $$ \begin{aligned} \min_{\mathbf{x}} && c_1x_1 + c_2x_2 \\ s.t. && a_{1,1}x_1 + a_{1,2}x_2 \geq b_1 \\ && a_{2,1}x_1 + a_{2,2}x_2 \leq b_2 \\ && a_{3,1}x_1 + a_{3,2}x_2 \leq b_3 \\ && a_{4,1}x_1 + a_{4,2}x_2 \geq b_4 \end{aligned} $$

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LINEAR PROGRAMMING

Linear objective with linear constraints

> $$ \begin{aligned} \min_{\mathbf{x}} && \mathbf{c}^T\mathbf{x} \\ s.t. && \mathbf{A}\mathbf{x} \leq \mathbf{b} \end{aligned} $$

As opposed to general optimization

> $$ \begin{aligned} \min_{\mathbf{x}} && f(\mathbf{x}) \\ s.t. && g_i(\mathbf{x}) \leq 0, i=1,\dots,M \\ && h_i(\mathbf{x}) = 0, i=1,\dots,P \end{aligned} $$

LINEAR PROGRAMMING

Different formulations

Inequality Form

$$ \begin{aligned} \min_{\mathbf{x}} && \mathbf{c}^T\mathbf{x} \\ s.t && \mathbf{A}\mathbf{x} \le \mathbf{b} \\ && ~~ \end{aligned} $$

General Form

$$ \begin{aligned} \min_{\mathbf{x}} && \mathbf{c}^T\mathbf{x} + d \\ s.t. && \mathbf{G}\mathbf{x} \leq \mathbf{h} \\ && \mathbf{A}\mathbf{x} = \mathbf{b} \\ \end{aligned} $

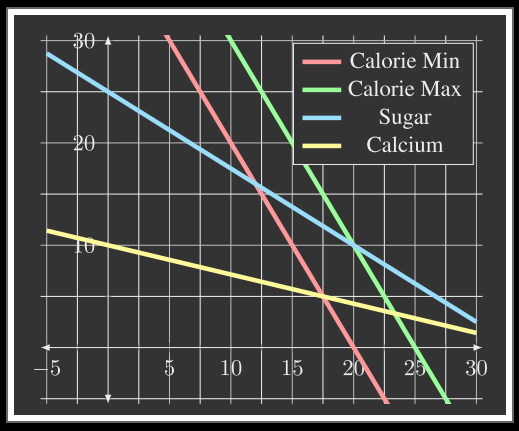
Standard Form (here)

$$ \begin{aligned} \min_{\mathbf{x}} && \mathbf{c}^T\mathbf{x} \\ s.t. && \mathbf{A}\mathbf{x} = \mathbf{b} \\ && \mathbf{x} \geq \mathbf{0} \end{aligned} $$

OPTIMIZATION

Problem Description

Graphical Representation



Optimization Representation

$$\begin{aligned} \min_{\mathbf{x}} && \mathbf{c}^T\mathbf{x} \\ \text{s.t.} && \mathbf{A}\mathbf{x} \leq \mathbf{b} \end{aligned}$$

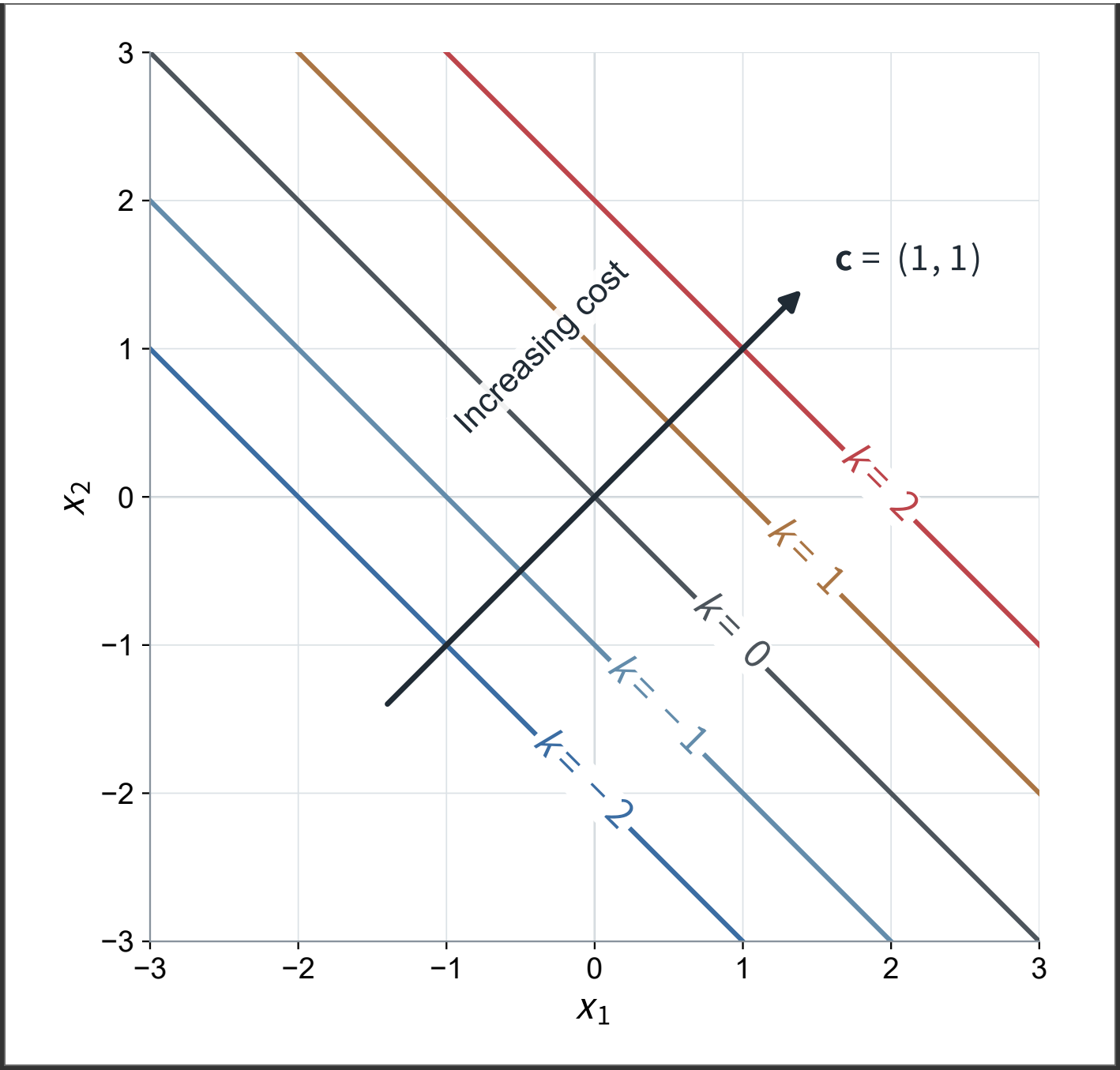
GRAPHICAL REPRESENTATION

What shape does this inequality represent?

$$ a_1x_1 + a_2x_2 \leq b_1 $$

COST CONTOURS

Illustration: $\mathbf{c}=[1,1]^T$.
Each line has constant cost:
$$\mathbf{c}^T\mathbf{x}=x_1+x_2=k.$$
$k\in\{-2,-1,0,1,2\}$.
The contours are parallel. The vector $\mathbf{c}$ is perpendicular to them and points toward **increasing cost**.


c = (1, 1)
Increasing cost
k = 2
k = 1
k = 0
k = −1
k = −2
x1
x2
3
2
1
0
−1
−2
−3
−3
−2
−1
0
1
2
3

LP GRAPHICAL REPRESENTATION

Inequality Form

$$ \begin{aligned} \min_{\mathbf{x}} && \mathbf{c}^T\mathbf{x} \\ s.t. && \mathbf{A}\mathbf{x} \leq \mathbf{b} \end{aligned} $$

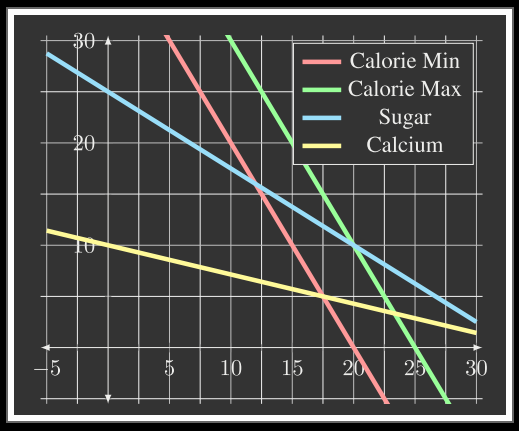
Start with no constraints

Add one constraint at a time

OPTIMIZATION

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Optimization Representation

$$\begin{aligned} \min_{\mathbf{x}} && \mathbf{c}^T\mathbf{x} \\ \text{s.t.} && \mathbf{A}\mathbf{x} \leq \mathbf{b} \end{aligned}$$

OPTIMIZATION FORMULATION

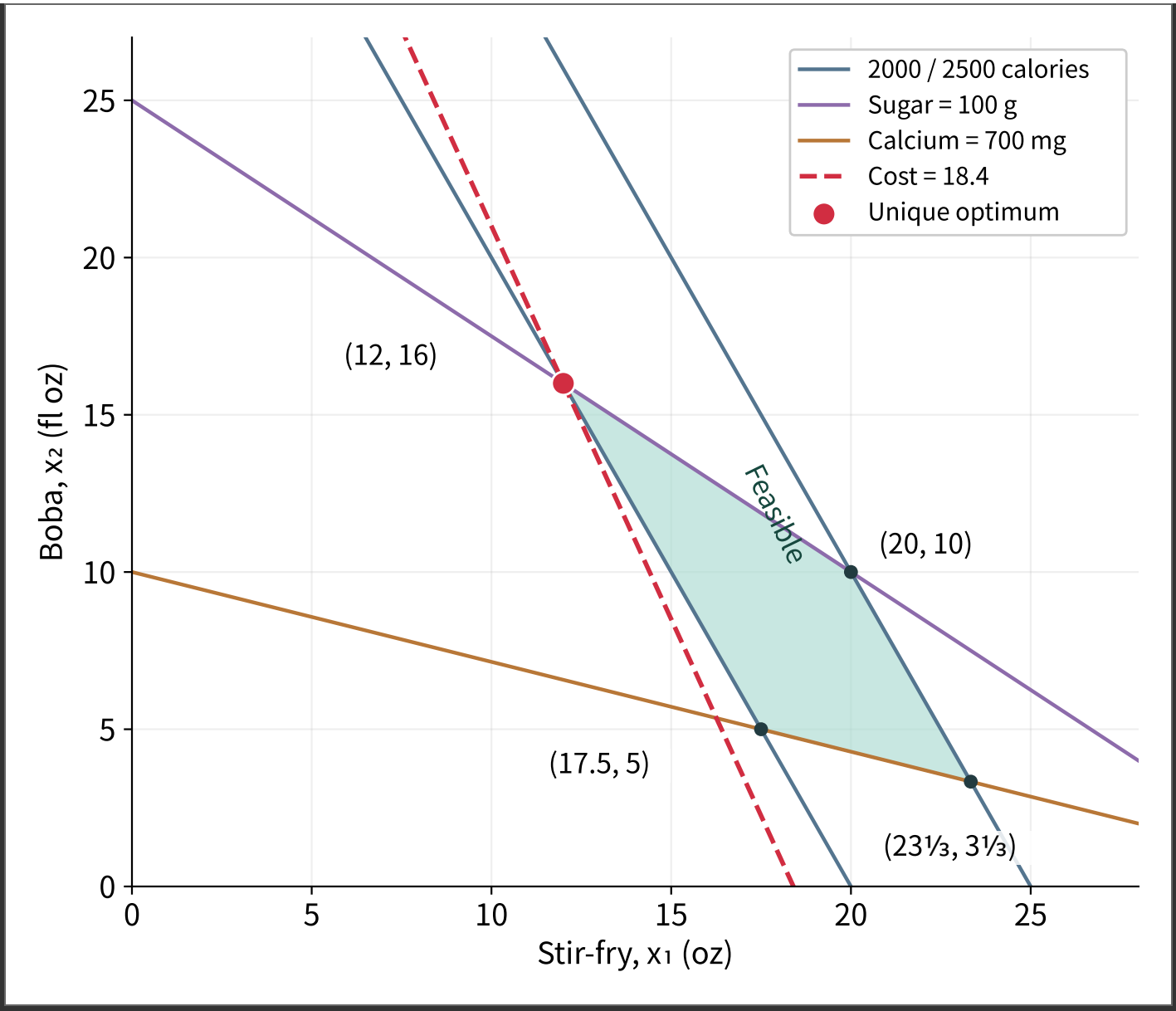
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SOLVING AN LP

For a nonempty, bounded feasible polyhedron, **at least one optimal solution is a vertex**.
For this diet model, the optimal vertex is unique.
In two dimensions, a vertex is a feasible intersection of independent constraint boundaries.


2000 / 2500 calories
Sugar = 100 g
Calcium = 700 mg
Cost = 18.4
Unique optimum
(12, 16)
Feasible
(20, 10)
(17.5, 5)
(23⅓, 3⅓)
Boba, x₂ (fl oz)
Stir-fry, x₁ (oz)
0
5
10
15
20
25

DIET PROBLEM: OPTIMAL SOLUTION

The calorie lower bound and sugar upper bound are binding:

$$\begin{aligned} 100x_1+50x_2&=2000,\\ 3x_1+4x_2&=100. \end{aligned}$$

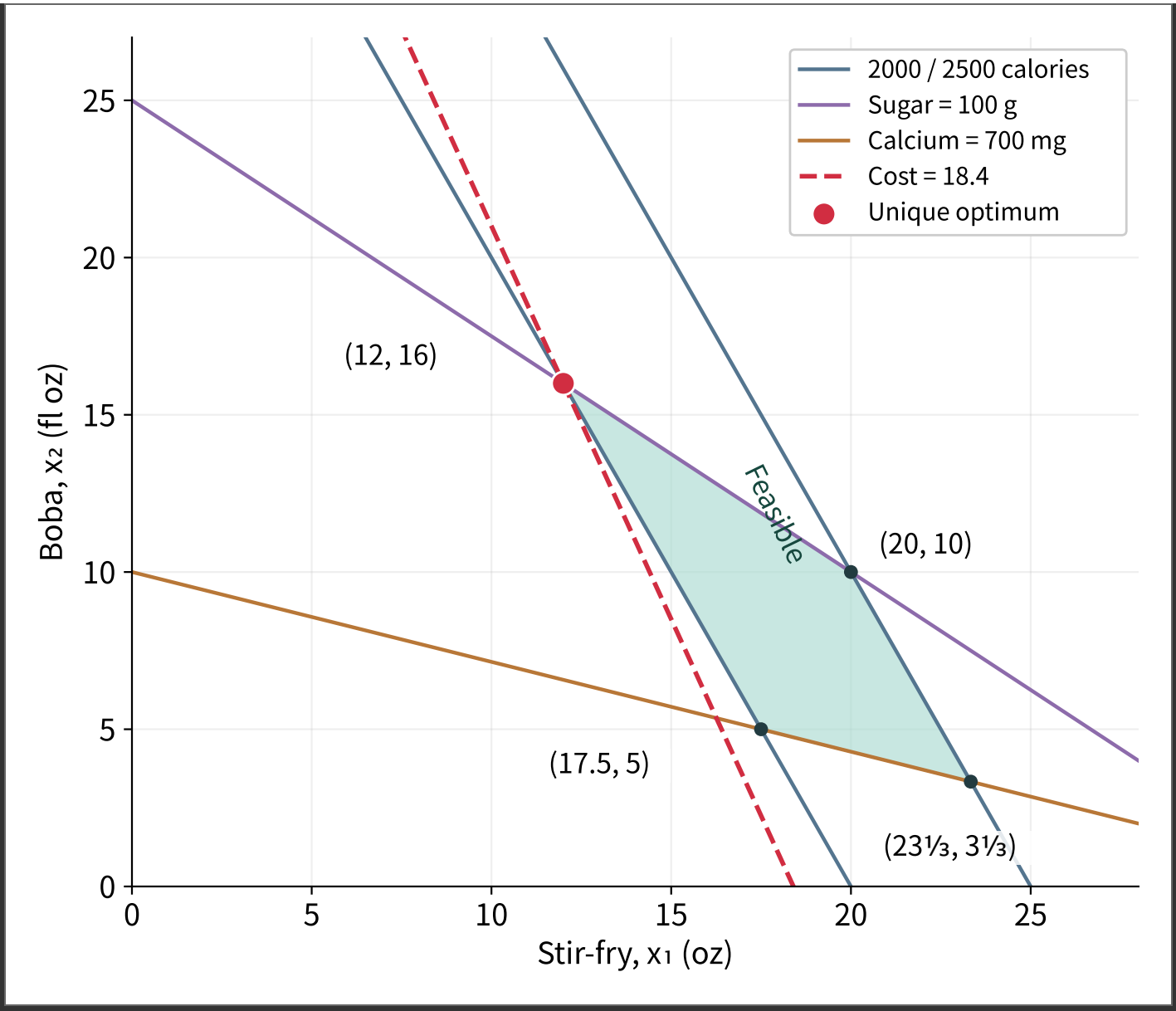
Unique optimum: $(x_1,x_2)=(12,16)$

12 oz stir-fry and 16 fl oz boba.

Minimum cost: 18.4 units

$12+0.4(16)=18.4$.

The other vertices cost 19.5, 24, and $24\frac{2}{3}$.


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POSSIBLE LP OUTCOMES

Outcome	Example	Meaning
Finite optimum	$\min x$ subject to $x\geq0$	Optimal value 0 at $x=0$
Infeasible	$\min x$ subject to $x\geq1$, $x\leq0$	No point satisfies all constraints
Unbounded objective	$\min -x$ subject to $x\geq0$	The objective can decrease without limit

An unbounded feasible region can still have a finite optimum, as in the first example.

ALGORITHMS AND INTEGER DECISIONS

Simplex: moves between adjacent feasible vertices, seeking a better objective value.

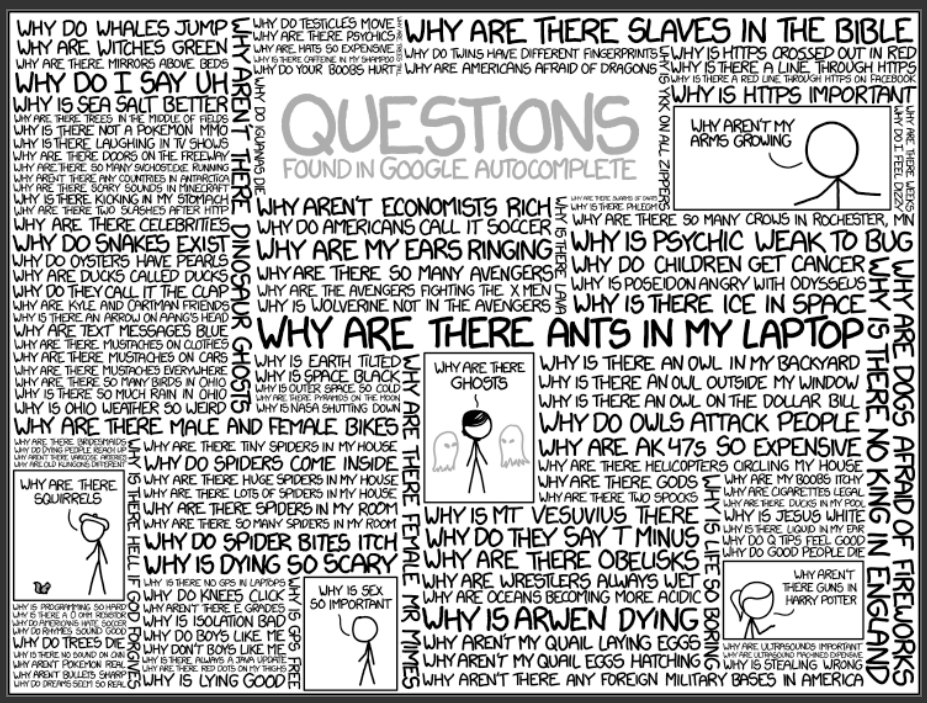
Interior-point methods: approach an optimum through the interior.

Requiring integer decisions gives an **integer program**. Rounding an LP solution may be infeasible or suboptimal.

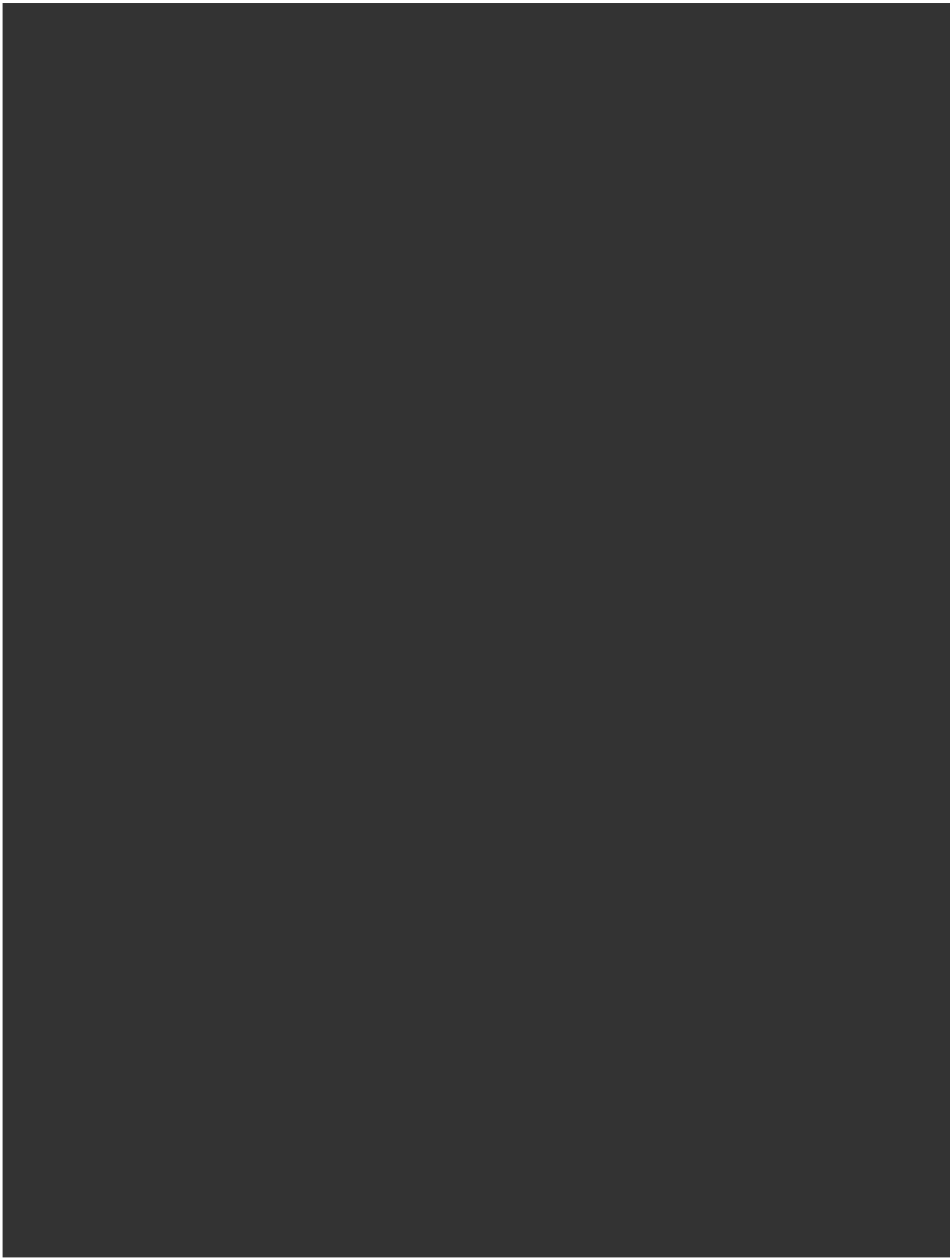
Branch and bound: uses LP relaxations to bound and prune a search tree.

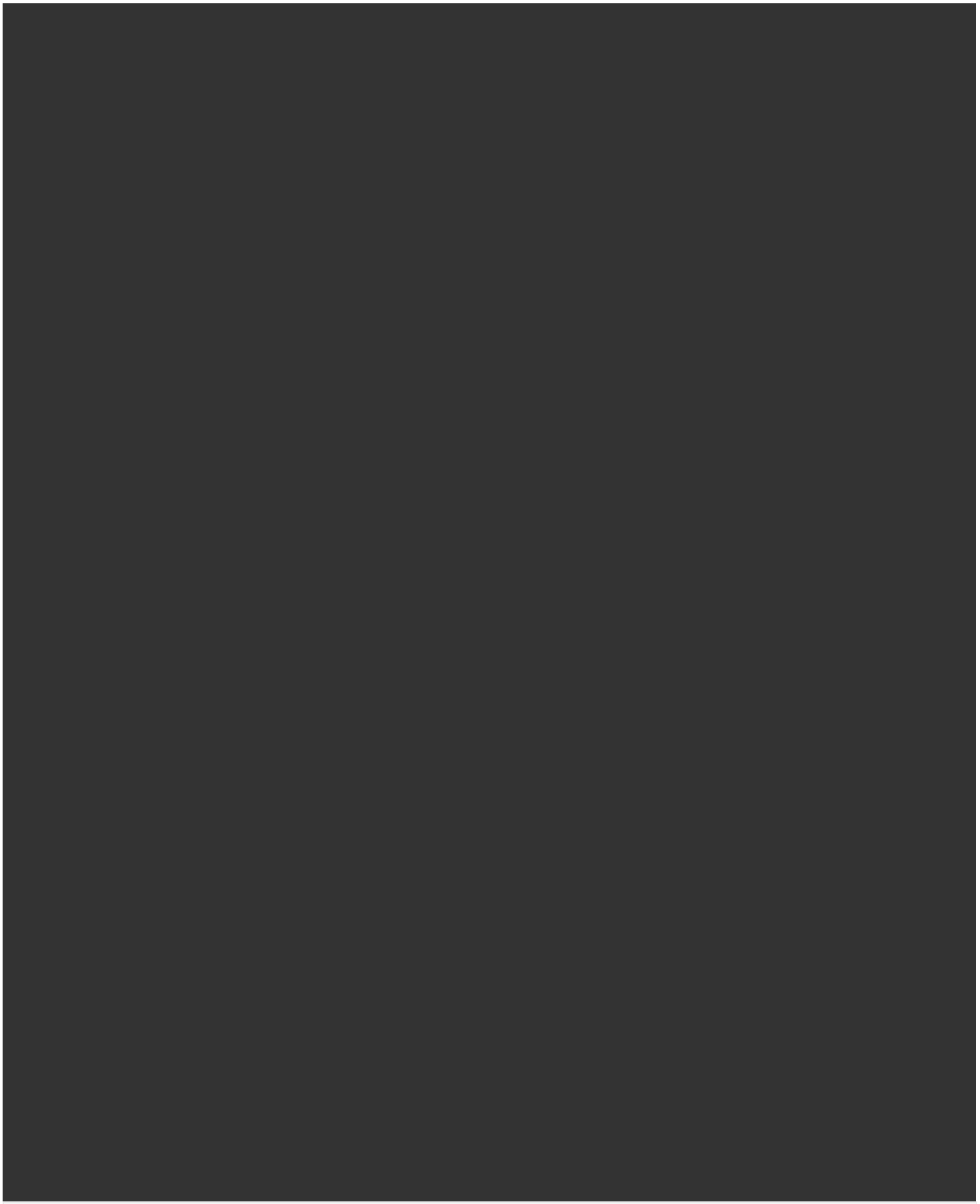
Open graphical LP and integer-programming demonstrations →

Q & A


QUESTIONS
FOUND IN GOOGLE AUTOCOMPLETE
WHY ARE THERE SLAVES IN THE BIBLE
WHY ARE THERE ANTS IN MY LAPTOP
WHY ARE THERE DINOSAUR GHOSTS
WHY ARE THERE FEMALE MR MIMES
WHY ARE DOGS AFRAID OF FIREWORKS
WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY ARE THERE SO MANY AVENGERS
WHY IS WOLVERINE NOT IN THE AVENGERS
WHY IS PSYCHIC WEAK TO BUG
WHY DO CHILDREN GET CANCER
WHY IS POSEIDON ANGRY WITH ODYSSEUS
WHY IS THERE ICE IN SPACE
WHY IS THERE NO KING IN ENGLAND
WHY IS HTTPS IMPORTANT
WHY AREN'T MY ARMS GROWING
WHY DO WHALES JUMP
WHY ARE WITCHES GREEN
WHY ARE THERE MIRRORS ABOVE BEDS
WHY DO I SAY UH
WHY IS SEA SALT BETTER
WHY ARE THERE TREES IN THE MIDDLE OF FIELDS
WHY IS THERE NOT A POKEMON MMO
WHY IS THERE LAUGHING IN TV SHOWS
WHY ARE THERE DOORS ON THE FREEWAY
WHY ARE THERE SO MANY SNOWSTORM RUNNING
WHY AREN'T THERE ANY COUNTRIES IN ANTARCTICA
WHY ARE THERE SCARY SOUNDS IN MINECRAFT
WHY IS THERE KICKING IN MY STOMACH
WHY ARE THERE TWO SUSHIES AFTER HTTP
WHY ARE THERE CELEBRITIES
WHY DO SNAKES EXIST
WHY DO OYSTERS HAVE PEARLS
WHY ARE DUCKS CALLED DUCKS
WHY DO THEY CALL IT THE CLAP
WHY ARE KYLE AND CARTMAN FRIENDS
WHY IS THERE AN ARROW ON AANG'S HEAD
WHY ARE TEXT MESSAGES BLUE
WHY ARE THERE MUSTACHES ON CLOTHES
WHY ARE THERE MUSTACHES ON CARS
WHY ARE THERE MUSTACHES EVERYWHERE
WHY ARE THERE SO MANY BIRDS IN OHIO
WHY IS THERE SO MUCH RAIN IN OHIO
WHY IS OHIO WEATHER SO WEIRD
WHY ARE THERE MALE AND FEMALE BIKES
WHY ARE THERE SQUIRRELS
WHY ARE THERE GHOSTS
WHY IS THERE AN OWL IN MY BACKYARD
WHY IS THERE AN OWL OUTSIDE MY WINDOW
WHY IS THERE AN OWL ON THE DOLLAR BILL
WHY DO OWLS ATTACK PEOPLE
WHY ARE AK 47s SO EXPENSIVE
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE
WHY ARE THERE GODS
WHY ARE THERE TWO SPOOKS
WHY IS MT VESUVIUS THERE
WHY DO THEY SAY T MINUS
WHY ARE THERE OBELISKS
WHY ARE WRESTLERS ALWAYS WET
WHY ARE OCEANS BECOMING MORE ACIDIC
WHY IS ARWEN DYING
WHY AREN'T MY QUAIL LAYING EGGS
WHY AREN'T MY QUAIL EGGS HATCHING
WHY AREN'T THERE ANY FOREIGN MILITARY BASES IN AMERICA
WHY ARE THERE TINY SPIDERS IN MY HOUSE
WHY DO SPIDERS COME INSIDE
WHY ARE THERE HUGE SPIDERS IN MY HOUSE
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE
WHY ARE THERE SPIDERS IN MY ROOM
WHY ARE THERE SO MANY SPIDERS IN MY ROOM
WHY DO SPIDER BITES ITCH
WHY IS DYING SO SCARY
WHY IS SEX SO IMPORTANT
WHY IS LIFE SO BORING
WHY ARE MY BOOBS ITCHY
WHY IS JESUS WHITE
WHY DO Q TIPS FEEL GOOD
WHY DO GOOD PEOPLE DIE
WHY AREN'T THERE GUNS IN HARRY POTTER
WHY DO TESTICLES MOVE
WHY ARE THERE PSYCHICS
WHY ARE HATS SO EXPENSIVE
WHY IS THERE CAFFEINE IN MY SHAMPOO
WHY DO YOUR BOOBS HURT
WHY DO TWINS HAVE DIFFERENT FINGERPRINTS
WHY ARE AMERICANS AFRAID OF DRAGONS
WHY IS HTTPS CROSSED OUT IN RED
WHY IS THERE A LINE THROUGH HTTPS
WHY IS THERE A RED LINE THROUGH HTTPS ON FACEBOOK
WHY DO I FEEL DIZZY
WHY ARE THERE WEEDS
WHY DO I SQUINT
WHY DO SQUIRRELS DIE
WHY AREN'T THERE
WHY IS THERE HELL IF GOD FORGIVES
WHY IS GPS FREE
WHY IS THERE LAVA
WHY IS THERE ENLIGHT
WHY IS THERE A TRX ON TAL ZIPPERS
WHY IS THERE LIQUID IN MY EAR
WHY ARE CIGARETTES LEGAL
WHY ARE THERE DUCKS IN MY POOL
WHY ARE ULTRASOUNDS IMPORTANT
WHY ARE ULTRASOUND PROBES DEPENDABLE
WHY IS STEALING WRONG
WHY IS THERE NO SOUND ON CNN
WHY AREN'T POKEMON REAL
WHY AREN'T BULLETS SHARP
WHY DO DREAMS SEEM SO REAL
WHY IS PROGRAMMING SO HARD
WHY IS THERE A 0 OHM RESISTOR
WHY DO AMERICANS HATE SOCCER
WHY DO RHYMES SOUND GOOD
WHY DO TREES DIE
WHY IS THERE NO GPS IN LAPTOPS
WHY DO KNEES CLICK
WHY AREN'T THERE E GRADES
WHY IS ISOLATION BAD
WHY DO BOYS LIKE ME
WHY DON'T BOYS LIKE ME
WHY IS THERE ALWAYS A JAVA UPDATE
WHY ARE THERE RED DOTS ON MY TABS
WHY IS LYING GOOD

XKCD





Speaker notes

